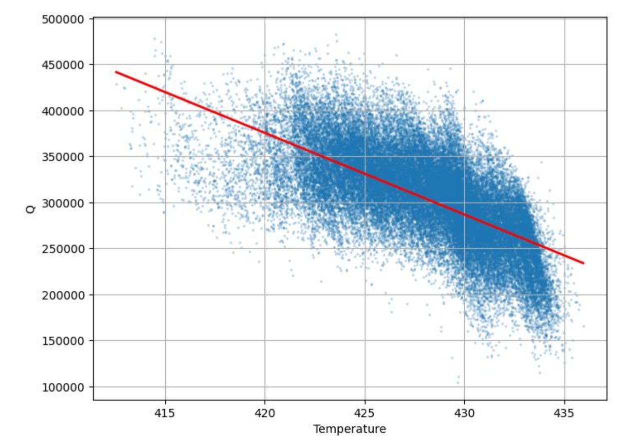
# Investigating the efficacy of data-driven techniques and Machine learning algorithms to predict heat transfer characteristics

*Raymond Twum-Barima - F231109 - Meng Mechanical Engineering - Wolfson school of Mechanical, Electrical and Manufacturing engineering*

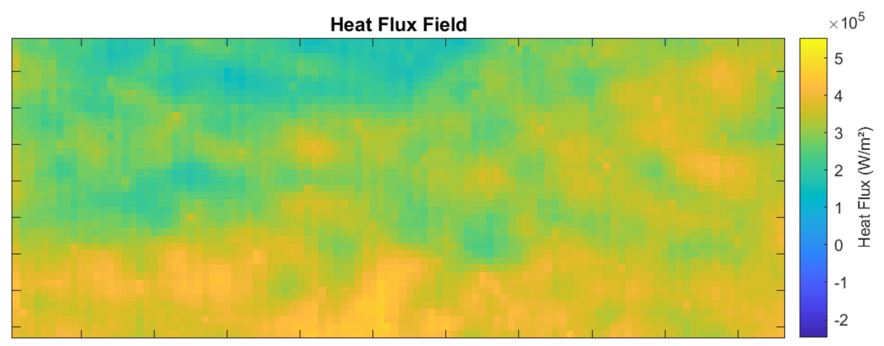
Loughborough University

## Section 1: Introduction



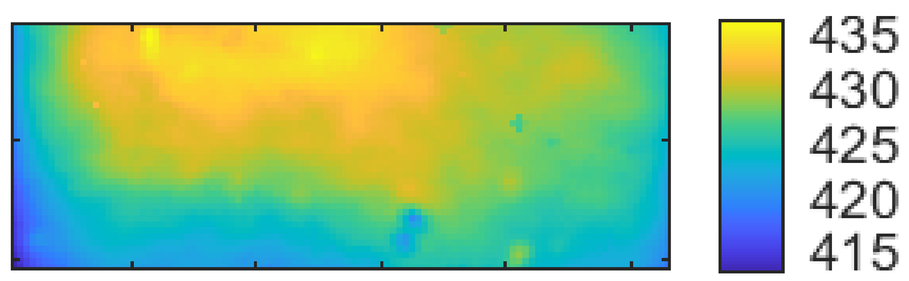
Scatterplot of Temperature and Heat Flux($R^2 = 0.4326$)

- Temperature explains some of the variation in heat flux




Raw Heat Flux Field

- More heterogeneous relative to heat flux
- Large range of distributions of values




Raw Temperature Field

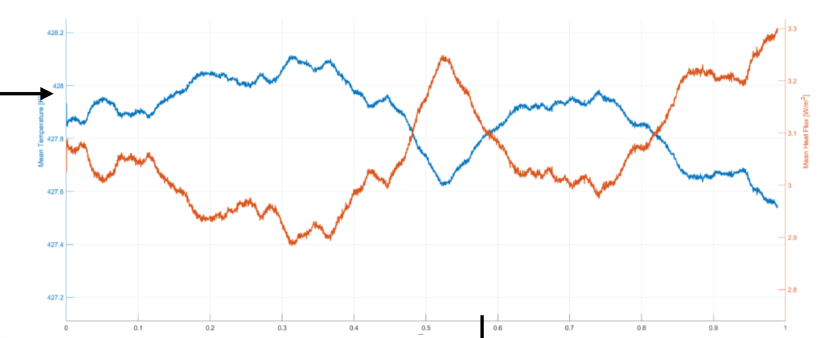
- More homogenous relative to heat flux
- Lower range of distributions of values

***Why this matters***

- Boiling Heat transfer is one of the most effective modes of thermal energy removal
- However, boiling remains difficult to predict due to its complex physics
- New advancements in Infrared Thermography have opened the door to data-driven and machine learning techniques to uncover this black box

## Section 2: Data processing and reduced order Modelling

**Raw IR Data** Q,T(x,y,z,t) → **Preprocessing** (removing edge effects and glitches) → **Data Validation**

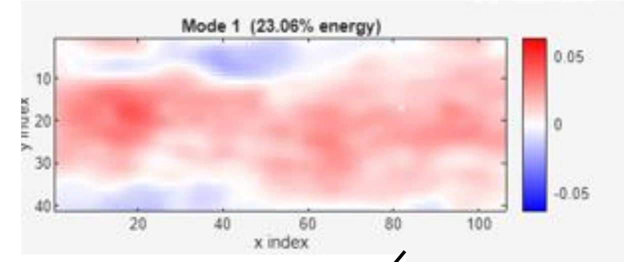


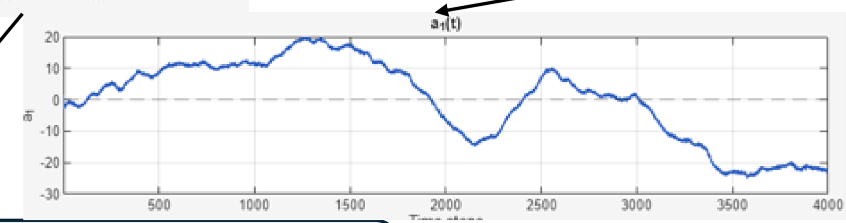
**Temperature data Matrix [41,106,4000]** 17,384,000 pixel time samples

Data split into training and test sets for ANN models 70:30

**Data is mean-centred** $X_c = X - X_{mean}$

**Proper Orthogonal Decomposition(POD)**



Top: spatial map of POD Temperature Mode 1, pattern within the fluctuation

Bottom: temporal coefficients of mode1. How strong is this pattern over time

**Dynamic Mode Decomposition (DMD)**

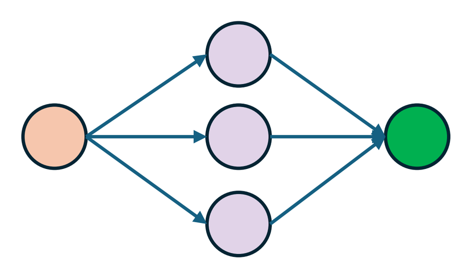
$$x(t) \approx \sum_{k=1}^{r} \phi_k \, a_k(t)$$

*Proper Orthogonal Decomposition (POD) is a reduced-order technique used in this project to identify the main patterns/modes within the heat flux and temperature Dataset*

## Section 3: Comparison of ANN models for heat-flux prediction

**ANN-Model A**

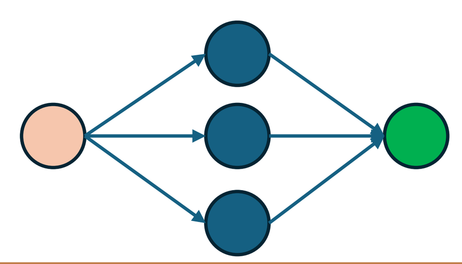
- Uses raw temperature only to predict local heat flux
- Acts as the baseline ANN model
- Shows temperature contains useful predictive information, but limited accuracy



Bayesian optimised hyperparameter
Neurons per layer : 32, Hidden layer: 2 ,Activation function: Relu

**ANN-Model B**

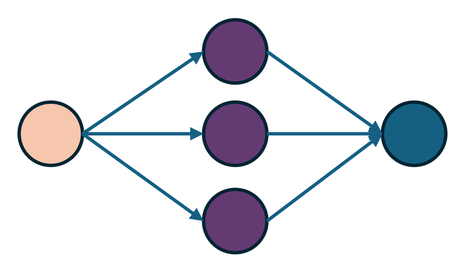
- Uses temperature and spatial gradient as inputs
- Motivated by adding physically meaningful local structure
- Only marginal improvement on model A, with signs of overfitting



Bayesian optimised hyperparameter
Neurons per layer : 64, Hidden layer: 3 ,Activation function: Tanh

**ANN-Model C**

- Uses the first five temperature POD modal contributions to predict the first five heat flux modal contributions, then reconstructs the temperature field
- Best Performing model



Bayesian optimised hyperparameter
Neurons per layer : 192, Hidden layer: 3 ,Activation function: Tanh

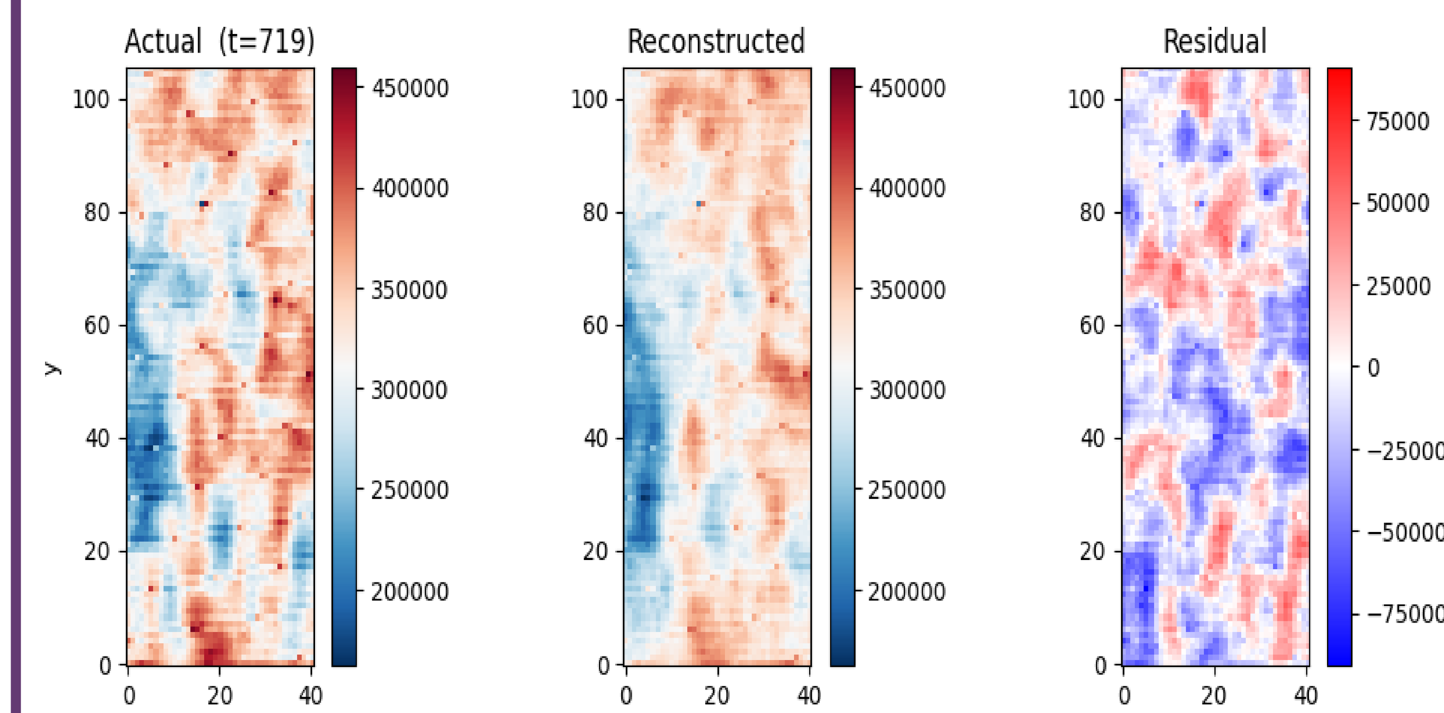
**Model Comparison Table**

| Test set error metrics | Linear baseline (T-q) | DMD | Model A | Model B | Model C |
|---|---|---|---|---|---|
| $R^2$ | 0.4326* | 0.3844 | 0.4887 | 0.4898 | **0.7293** |
| RMSE ($W/m^2$) | - | 39144.27 7 | 35676.20 2 | 35638.62 7 | **25958.51 7** |
| MAE ($W/m^2$) | - | 30525.38 6 | 28200.55 5 | 27791.53 9 | **20655.79 2** |

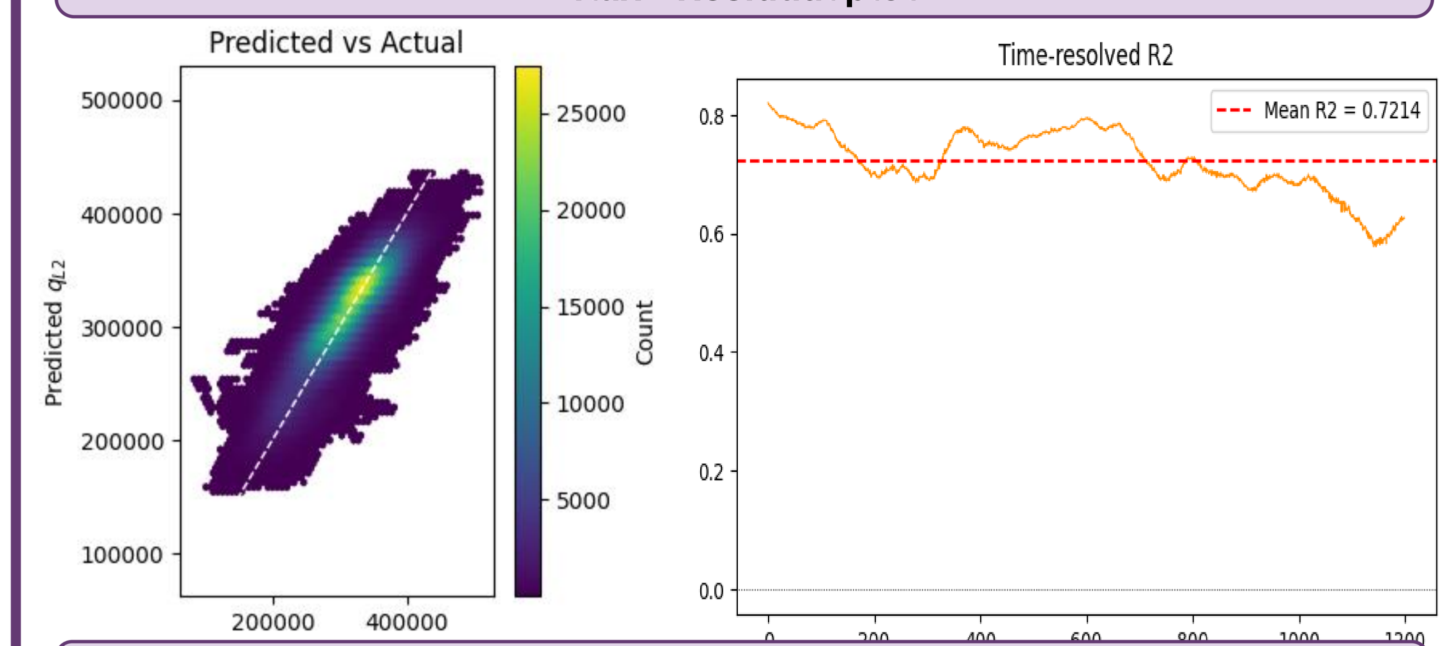
**Key Results:** Model C outperformed the linear baseline and the raw feature ANN models, showing that low-order POD structure captures the dominant temperature heat flux relationship more effectively

An ANN is a computational model loosely inspired by the brain. It learns patterns from data by passing inputs through layers of interconnected nodes ("neurons"), each applying a weighted transformation. $w_{new} = w_{old} - \eta \times \nabla L$

## Section 4: Key Results figure




**Left to right (test set): true heat flux plot – Reconstructed/predicted heat flux – Residual plot**




**Left to right: predicted vs Actual Heat Flux scatterplot – Time-resolved $R^2$**

## Section 5: Conclusion

- Model C (POD modal mapping) achieved $R^2 = 0.729$ — a 69% improvement over the linear baseline, outperforming all other methods
- Raw temperature inputs (Models A & B) plateaued at $R^2 \approx 0.49$, confirming that low-dimensional modal features capture the heat flux relationship more effectively than raw inputs
- Future work should explore more POD modes, wider datasets and improved regularisation to further improve predictive accuracy

## Acknowledgments

- My supervisor, who kept me going the whole way: Dr Huayong Zhao
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- Professor J. Nathan Kutz, whose YouTube videos demystified the POD and DMD